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BHATTADEV UNIVERSITY

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(ECONOMICS)

TITLE OF THE PROJECT REPORT

**Comparative Study of Traditional vs. Modern Dairy
Farming Practices**
PAPER: ECO-HE-6026-THE ECONOMY OF ASSAM

SUBMITTED BY

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6TH SEM, B.A (ECONOMICS)
REGISTRATION NO: 2210109
SESSION: 2022-2025

UNDER THE SUPERVISION

MRS RITU BARUA
DEPARTMENT OF ECONOMICS

UNDERTAKING THE PROJECT REPORT
ON
BACHELOR OF ARTS (ECONOMICS)

TOPIC OF THE PROJECT REPORT

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DEGREE – BACHELOR OF ARTS (ECONOMICS).

I hereby declare that above this work has been done by me and has not
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CONTENT

CHAPTERS	PAGE NO.
ABSTRACT	1
CHAPTER 1: Introduction 1.1 Overview of Dairy Production 1.2 Aim of Study 1.3 Area of study	2 - 3
CHAPTER 2: Literature review of the study 2.1 Research Gap 2.2 Objectives of the study	4 - 9
CHAPTER 3: Methodology of the study 3.1 Data Collection 3.2 Significance of the study area 3.3 Sample design 3.4 Methodological tools used in analysis	10 - 13
CHAPTER 4: Analysis the Traditional and Modern dairy farming 4.1 Traditional vs Modern output 4.2 Sustainability Indicators 4.3 Livelihood outcomes 4.4 Animal welfare comparison 4.5 Adoption Patterns and Influencing Factors	14 - 22
CHAPTER 5: Discussion 5.1 Interpretation of research 5.2 Comparative advantages and disadvantages 5.3 Socio-economics implications	23 - 25
CHAPTER 6: Conclusion 6.1 Future research 6.2 Limitations 6.3 Importance of the study 6.4 Key finding recap	26 - 28
References	29 - 30
Appendix 1	31 - 32
Appendix 2	33 - 35

ABSTRACT

Dairy farming is a significant livelihood activity in rural Assam, especially in the Bajali district, where both traditional and modern practices coexist. This study focuses on two Gaon Panchayats—Manikpur and Bandhasidhani—to conduct a comparative analysis of traditional and modern dairy farming systems. Data collected from 38 dairy farmers highlights a predominant reliance on mixed practices that combine age-old methods with emerging modern techniques.

The research evaluates various aspects including milk yield, income generation, animal health, sustainability, and access to veterinary care. It also considers the role of tribal and non-tribal farmers, as well as the active participation of women in dairy operations. Findings suggest that farmers adopting improved breeds, scientific feeding, and better management practices report higher productivity and income. However, challenges such as limited veterinary access, poor infrastructure, and financial constraints continue to hinder progress.

This study concludes that integrating traditional knowledge with modern techniques—supported by training, infrastructure development, and policy support—can make dairy farming more sustainable and economically viable for rural communities in Assam.

INTRODUCTION

1.1 Overview of Dairy Production

Dairy farming plays a vital role in the rural economy of Assam, particularly in the Bajali Area, where it serves as a primary livelihood for many households. Traditionally, dairy farming in this region has been characterized by conventional methods involving indigenous cattle breeds, manual milking, and basic feeding practices. However, with the advent of modern agricultural techniques, there has been a gradual shift towards more efficient, technology-driven dairy farming practices.

Modern dairy farming practices in Bajali areas include the use of crossbred cattle, improved feeding techniques, and scientific herd management. These practices have significantly impacted milk productivity, animal health, and farmers' economic conditions. In contrast, traditional methods, despite being less efficient, hold cultural significance and require lower initial investments.

This comparative study aims to analyse the differences between traditional and modern dairy farming practices in the Bajali area of Assam. It will evaluate the economic, social, and environmental impacts of both approaches while highlighting the challenges and opportunities associated with their adoption. Through this study, we seek to provide insights into how modernization affects dairy farming and the potential for sustainable development in the region.

1.2 Aim of Study

The primary aim of this study is to conduct a comprehensive comparison between traditional and modern dairy farming practices in the Bajali region of Assam. The study intends to examine how each approach influences milk production, income generation, animal welfare, and sustainability. By exploring the economic, technological, and social dimensions of dairy farming, the research seeks to identify the strengths and limitations of both traditional and modern methods. Additionally, this study aims to assess how modernization is reshaping the rural dairy sector and to offer practical recommendations for improving dairy productivity and livelihoods while preserving valuable traditional knowledge.

1.3 Area of Study

The study was conducted exclusively in two Gaon Panchayats—Manikpur and Bandhasidhani—situated in the Bajali district of Assam. Bajali, a newly formed district carved out of Barpeta in 2021, is known for its agrarian economy with dairy farming serving as a key livelihood activity. Although the region as a whole reflects a mix of traditional and modern dairy practices, this study focuses specifically on the selected Gaon Panchayats, where both indigenous methods and elements of modern dairy techniques are observed. These areas offer valuable insights into the evolving nature of rural dairy farming within a defined and realistic setting.

Literature review of the study

A literature review provides an overview of the existing research on a particular topic. It helps to identify what is already known, what gaps exist, and how the current study fits into the broader academic conversation. In this review, I will summarize and analyze key studies related to "The Comparative Study of Traditional Vs Modern dairy farming practices" highlighting major findings, ongoing debates, and areas where further research is needed.

1. **Das (2014)**, in their research paper titled "**Impact of Dairy Farming for Changing Livelihood through AACP Intervention in Bajali Area of Barpeta District, Assam,**" explored how dairy farming influenced rural livelihoods in the Bajali region during the implementation of the Assam Agricultural Competitiveness Project (AACP). The study emphasizes that dairy farming, as part of the broader livestock sector, holds significant potential for sustaining rural livelihoods in developing countries like India. Assam, located in the North Eastern Region, is identified as one of the more economically backward states in the country, with notably low per capita availability of livestock products such as milk, eggs, and meat. To meet local demand, the state heavily relies on imports from other parts of India. However, during the AACP intervention, there was notable progress in dairy development, particularly in and around the Bajali Development Block of Barpeta District. This development was marked by the formation of Dairy Cooperative Societies, which played a critical role in increasing milk production. The findings of the study reveal that the intervention had a significant socio-economic impact on the livelihoods of dairy farmers in the area. The authors conclude that, if challenges faced by the farmers are adequately addressed, dairy farming has the potential to reduce poverty and contribute to a sustainable rural livelihood system in the long run.
2. **Sarma and Barman (2014)**, in their study titled "**Supply Chain Analysis in Milk Marketing: A Comparative Study of DUSS Member and Non-Member Dairy Farmers in Barpeta District of Assam,**" explored the functioning of milk marketing systems in both organized and unorganized

sectors of the dairy supply chain in Assam. The study emphasized that milk marketing in the state is predominantly unorganized, with over 90% of unprocessed milk being traded through informal channels. However, in select regions such as Barpeta district, cooperative societies like the Dugdha Unnayan Samabai Sammittees (DUSS) have contributed to structuring milk supply networks. The research focused on the Bajali and Bhawanipur development blocks of the district and aimed to evaluate the marketing patterns, costs, margins, price spread, and efficiency across different milk supply chains. Data from the year 2012–13 were collected through questionnaires and surveys from member and non-member dairy farmers.

The findings revealed that local cattle breeds were more common on small farms, while crossbred Jersey cows were preferred across all farm sizes due to their higher milk yield and better fat and SNF content. The marketing behavior also differed by group: non-member small farmers mostly sold milk directly to consumers, while medium and large non-member farmers distributed milk through vendors, retailers, and restaurants. In contrast, DUSS member farmers sold their milk to their respective cooperative societies, which then supplied processed milk to organized outlets such as Purabi Dairy (WAMUL) in Guwahati and Asomi Dairy in Sarbhog.

The study identified five marketing channels for non-members and two for DUSS members, highlighting a more streamlined and efficient supply chain within the cooperative model. Furthermore, the employment generated by DUSS was substantial, with an average of over 1,800 man-days per year in Bajali block and over 1,200 man-days in Bhawanipur block, combining both full-time and part-time opportunities. Overall, the study concluded that DUSS significantly improved milk marketing efficiency, income distribution, and employment generation in the region.

3. **Das (2014)**, in his study titled “**Dairy Cattle Management and Marketing of Milk in Lower Brahmaputra Valley Zone of Assam,**” conducted an in-depth investigation into dairy farming practices by surveying 200 randomly selected dairy farmers, each owning at least two milch cows. Using a pre-tested and structured interview schedule, data were collected and analyzed through

standard statistical methods to draw meaningful conclusions aligned with the study's objectives. The research focused on key economic traits of dairy cattle such as age at first calving, lactation yield, lactation length, dry period, service period, and intercooling period, comparing performance between crossbred and indigenous cattle. The results revealed that crossbred cattle significantly outperformed indigenous breeds across all traits. For instance, the average lactation yield in crossbred cattle was recorded at 1441.18 litres, compared to only 280.75 litres in indigenous breeds. Similarly, crossbred cattle showed earlier calving (31.03 months) than indigenous ones (42.34 months), along with shorter service and dry periods.

The study also examined housing conditions, noting that 56.84% of crossbred cattle were housed under thatched roofs and 43.16% under corrugated iron sheets. In contrast, 81.48% of indigenous cattle were kept under thatched roofing. Regarding wall types in cattle sheds, the study found that for crossbred cattle, 47.95% of farmers used sheds with half walls, 23.97% used full walls, and 28.08% had no walls. For indigenous cattle, a majority (62.96%) used full-walled sheds. The findings showed that both genetic group and location significantly influenced economic traits, except for lactation yield in indigenous cattle across locations. This study highlights the superior performance of crossbred cattle in terms of productivity and underscores the need for improved housing infrastructure to support dairy development in the region.

4. **Kakaty and Das (2017)**, in their study titled “**Assessment of the Status of Dairying and Potential to Improve Socio-Economic Status of the Milk Producers and Convergence of all Central and State Schemes at District Level in Assam,**” examined the current condition of the dairy sector and its potential to uplift rural livelihoods across four districts of Assam—Barpeta, Jorhat, Kamrup, and Nagaon. The research involved 240 milk producers, comprising 120 members from Dairy Cooperative Societies (DCS) and 120 non-members (NDCS), aiming to assess both the operational status of dairying and the effectiveness of government interventions. The authors highlight the multifaceted role

of dairy farming in rural India, noting its contribution not only to employment and income generation but also to nutritional security and organic manure production through cow dung, thereby enhancing soil fertility and crop productivity.

Despite Assam's natural resources and favorable conditions for dairy development, the study found that the sector remains underdeveloped due to numerous structural and infrastructural challenges. These include inadequate healthcare services for livestock, limited access to improved breeding techniques like Artificial Insemination, poor milk marketing systems, and gaps in timely vaccination. The study concluded that convergence and effective implementation of central and state schemes at the district level, along with optimal resource utilization and improved support services, could significantly improve the socio-economic status of milk producers in Assam.

5. **Tamuli et al. (2021)**, in their research titled “**Attributes Influencing Adoption of Health Care Scientific Dairy Management Practices among Tribal and Non-Tribal Dairy Farmers of Assam,**” investigated the key variables that influence the adoption of scientific health care practices in dairy management across tribal and non-tribal communities in Assam. Using a descriptive survey design, the study was conducted over a period of six months—from October 2019 to March 2020—in selected tribal and non-tribal villages of Sonitpur, Lakhimpur, and Dhemaji districts. A total of 240 respondents (120 tribal and 120 non-tribal dairy farmers) were personally interviewed using a structured interview schedule to collect relevant data.

The findings revealed that several attributes—including prior experience in dairy farming, level of extension contact, attitude toward dairy farming, and aspiration level—played a significant role in determining the extent of adoption of scientific dairy health care practices. Notably, these factors explained approximately 71% of the variation in adoption levels among tribal farmers. The study highlights the importance of socio-psychological and institutional variables in bridging the gap between traditional and scientific dairy practices, particularly in tribal areas where access to formal training and veterinary services may be limited. The research emphasizes the need for targeted

interventions, awareness programs, and customized training modules to promote scientific dairy practices among different community groups in Assam.

6. **Swargeary and Kalita (2023)**, in their study titled “**Study on Socio-Economic Status of Dairy Farmers and Its Impact on Environment at Topatoli Village of Kamrup Metropolitan District of Assam, India,**” examined both the economic conditions of dairy farmers and the environmental implications of dairy practices in a rural setting. Conducted in Topatoli village of Kamrup (M) district, the study used a mixed-method approach that combined structured questionnaires with face-to-face, in-depth interviews to gather both quantitative and qualitative data. Out of the 21 respondents, 90.48% identified dairy farming as their primary and sole livelihood source, reflecting its central role in sustaining rural households.

The study also revealed considerable income variation among the farmers. While only 14.28% earned more than ₹50,000 per month, a larger share (52.38%) fell within the income bracket of ₹11,000–₹20,000 per month. Income groups between ₹21,000–₹30,000 and ₹31,000–₹40,000 each constituted 9.52% of respondents. These findings underline the modest economic status of most dairy farmers in the region. Though the study primarily focused on socio-economic dimensions, it also aimed to assess the environmental implications of dairy activities, recognizing that sustainable dairy development must consider both livelihood and ecological factors. The use of both qualitative and quantitative tools added depth to the understanding of how dairy farming influences rural livelihoods and local ecosystems.

2.1 Research Gap

While several studies have explored various aspects of dairy farming in Assam—such as socio-economic impacts, marketing efficiency, housing practices, breed performance, and health management—there is a notable lack of comparative analysis between traditional and modern dairy farming practices in terms of sustainability, productivity, and adoption barriers. No study

comprehensively evaluates how these two systems differ in practical outcomes, especially across diverse socio-economic and geographic contexts in Assam. This leaves a gap for research that directly compares the two to inform policy and practice.

2.2 Objectives of the study

- To compare the socio-economic impacts of traditional and modern dairy farming practices on rural livelihoods in the selected Gaon Panchayats of Bajali district, Assam.
 - To assess the adoption level and influencing factors of scientific dairy management practices among tribal and non-tribal farmers in Assam.
-

Methodology of the Study

3.1 Data Collection

This study is based on both **primary** and **secondary data** sources to ensure a comprehensive and reliable analysis of traditional and modern dairy farming practices in the Bajali district (Manikpur and Bandhasidhani Gaon panchayat) of Assam.

Primary Data:

Primary data was collected directly from dairy farmers in two panchayats of the Bajali district through a structured questionnaire. The survey included both open-ended and close-ended questions focusing on key aspects such as:

- Type of farming practice (traditional/modern)
- Cattle breeds and milk yield
- Feeding methods and animal healthcare
- Milking and storage systems
- Marketing and income patterns
- Challenges faced by farmers

Respondents were selected using a **random sampling method**, ensuring representation from both traditional and modern farming households. The data was gathered through **personal interviews and field visits**.

Secondary Data:

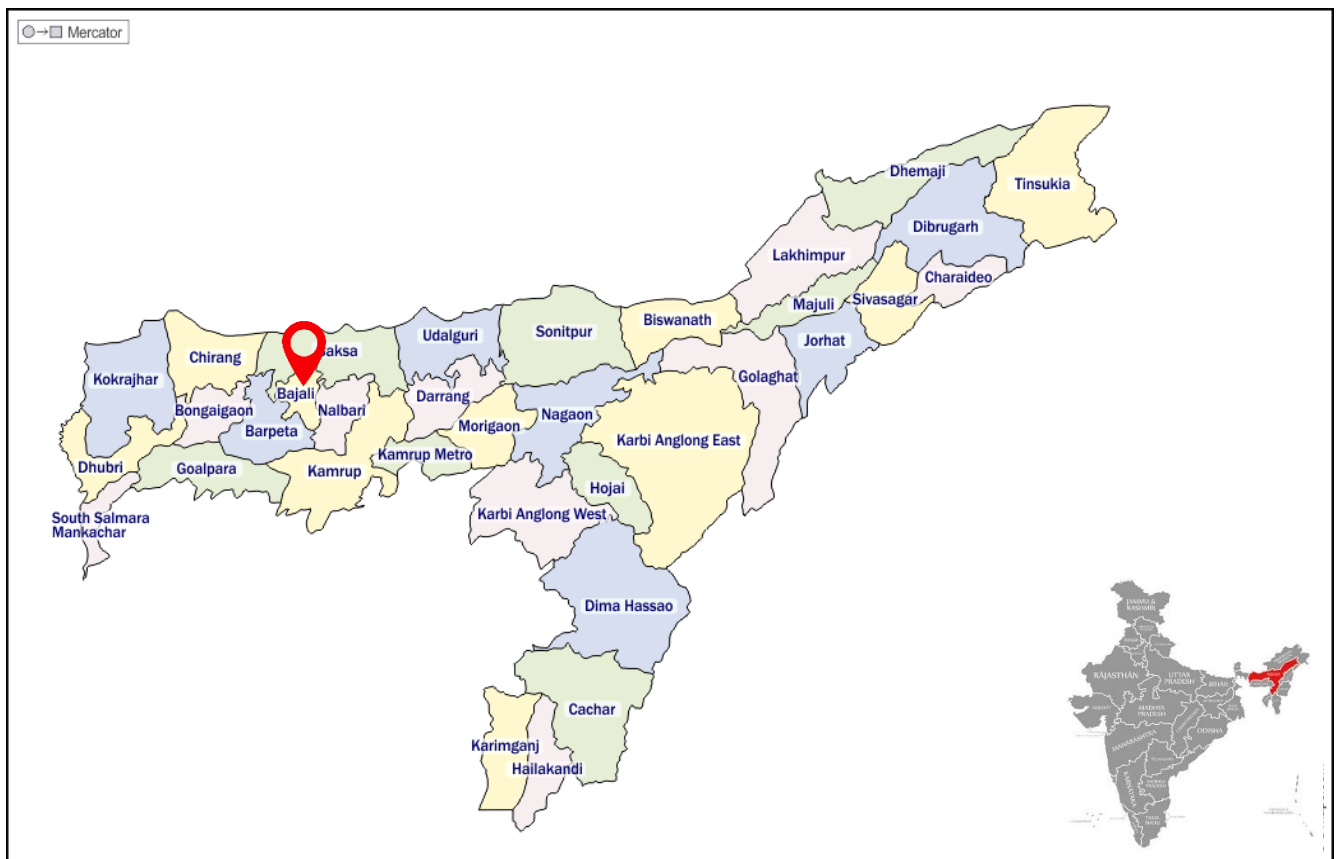
Secondary data was gathered from credible sources such as:

- Research journals
 - Previous studies on dairy farming in Assam and India
 - Various articles from credible sources
-

These sources helped in understanding broader trends, cross-verifying primary data, and providing context for the comparative analysis.

3.2 Significance of the study area

The Gaon Panchayats, **Manikpur** and **Bandhasidhani** in Bajali district were selected for this study due to their strong reliance on agriculture and dairy farming as key sources of livelihood. Many households in these areas are engaged in small-scale dairy activities, making them suitable for examining both traditional and modern dairy practices side by side. These Panchayats reflect a transitional phase—while some farmers continue with traditional methods, others are gradually adopting modern approaches such as improved cattle breeds and scientific management techniques.



Studying the Gaon Panchayats, Manikpur and Bandhasidhani in Bajali district offers valuable insights into how modernization is shaping rural dairy farming in Assam. It also helps to understand the challenges faced by traditional farmers in adopting new technologies, as well as the advantages experienced by those who have begun transitioning to modern practices.

3.3 Sample design

To make the study more accurate and meaningful, efforts were made to include both traditional and modern dairy farmers from two Gaon panchayat of Bajali district, Assam. The respondents were selected using a **random sampling method**, which helped avoid bias and allowed the findings to better reflect the real situation in the area.

A total of 38 dairy farmers were surveyed for this study. The majority of them follow a combination of both traditional and modern practices, blending age-old methods with newer techniques to suit their needs and resources. Only a few farmers continue to rely entirely on traditional methods, while a very small number have adopted fully modern systems. This mix of practices provided a rich and realistic understanding of how dairy farming is evolving in the rural area of Bajali district. Farmers were selected from various villages and localities falling under the jurisdiction of the Manikpur and Bandhasidhani Gaon Panchayats. Care was taken to include farmers from different age groups, levels of experience, and farm sizes so that the study could capture a wide range of opinions and practices.

Additionally, the study aimed to assess the **adoption level and the influencing factors of scientific dairy management practices among tribal and non-tribal farmers** in the selected areas. By examining how these different communities' approach dairy farming, the research provides insights into the social, cultural, and economic factors that shape the transition toward modern dairy practices. This approach helped the research compare the practical, economic, and technical differences between traditional and modern dairy farming in a real-world setting. It also ensured that both small-scale farmers and those using newer, advanced methods were fairly represented in the study.

3.4 Methodological tools used in analysis

To study and compare traditional and modern dairy farming practices, some basic tools were used to help organize and understand the information collected from farmers. These tools made it easier to analyse the responses and present the results clearly.

1. Percentage Method:

This was used to calculate and show the number of farmers using different practices (like feeding methods, cattle breeds, milking techniques, etc.). It helped show the data in a simple and understandable way.

2. Tables and Charts:

Simple tables and pie/bar charts were used to compare the results between traditional, modern, and mixed farming systems. This helped make the differences clear and easy to read.

3. Descriptive Analysis:

Some questions were open-ended, where farmers gave their opinions or explained their experiences. These answers were explained in simple words (descriptive form) to show what farmers think or feel.

4. Observation:

During the field visits, observations were made about cattle sheds, feeding systems, and milking processes. These were used to support the answers given by farmers.

These simple tools helped analyse both numbers and opinions, making the study easier to understand and present.

Analysis the Traditional and Modern dairy farming

4.1 Traditional vs. Modern Output

Based on the 38 responses collected from dairy farmers in the Bajali district—from Manikpur and Bandhasidhani Gaon Panchayats—it was observed that the majority of the farmers, including both tribal and non-tribal groups, practiced a mixed system that blends traditional methods with modern dairy techniques. Only a few respondents, mostly from tribal backgrounds, relied entirely on traditional methods, while none of the farmers had fully adopted a completely modern dairy system.

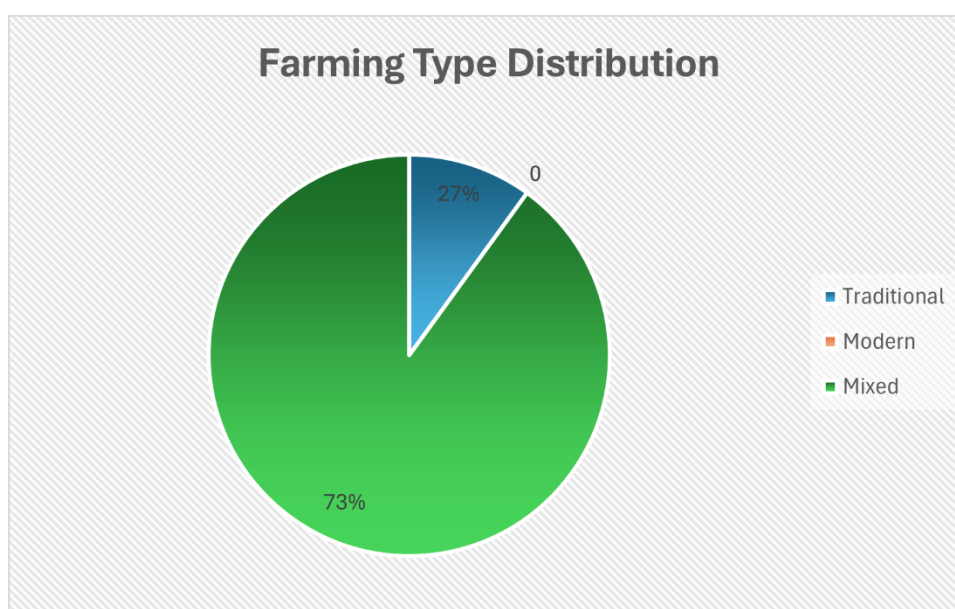


Fig 1: Type of Farming Practices among Dairy Farmers

This chart illustrates that approximately 73% of the farmers follow a mixed system. A smaller proportion still adheres solely to traditional methods, and no respondents reported complete adoption of modern systems.

Milk yield per cow per day was significantly influenced by the type of cattle breed and feeding method. Farmers who used crossbred or exotic cattle along with mineral supplements and commercial feed reported higher yields (around 10–15 litres/day). In contrast, farmers depending solely on indigenous breeds, particularly those following traditional methods, reported lower yields

(approximately 5–8 litres/day). This trend was common among both tribal and non-tribal farmers, though non-tribal farmers demonstrated relatively higher adoption of improved cattle breeds and scientific feeding practices.

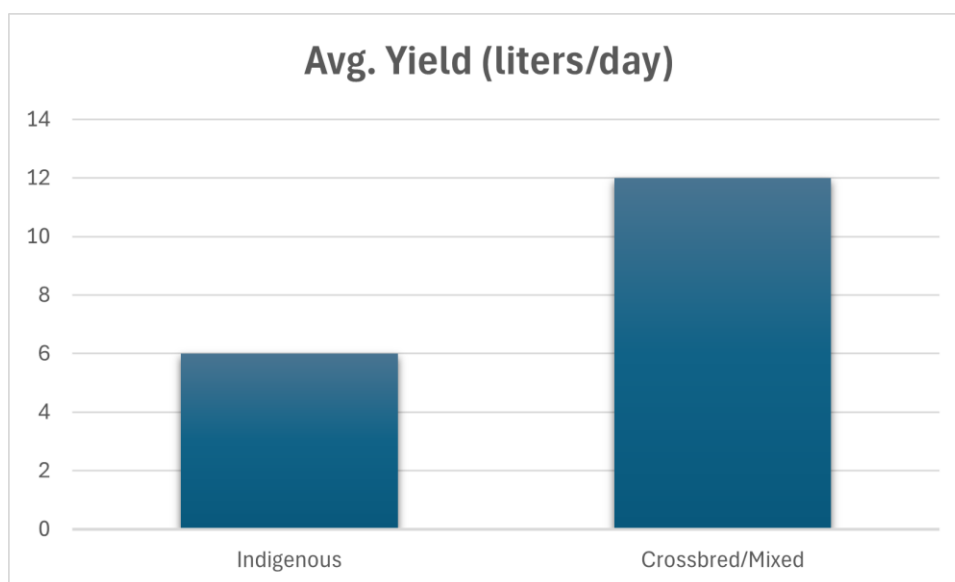


Fig. 2: Average Milk Yield by Breed Type

This figure shows that crossbred cattle produce more milk than indigenous breeds, reaffirming the benefits of modern breeding practices.

Feeding practices also reflected varying levels of adoption. Most farmers—regardless of community—used a mix of natural grass, cultivated fodder, and commercial feed. However, a deeper look shows that non-tribal farmers had slightly higher awareness and application of scientific feeding, often influenced by access to training, market exposure, and financial capacity. While some tribal farmers were aware of such practices, their adoption remained comparatively lower, often due to lack of extension support and resource constraints.

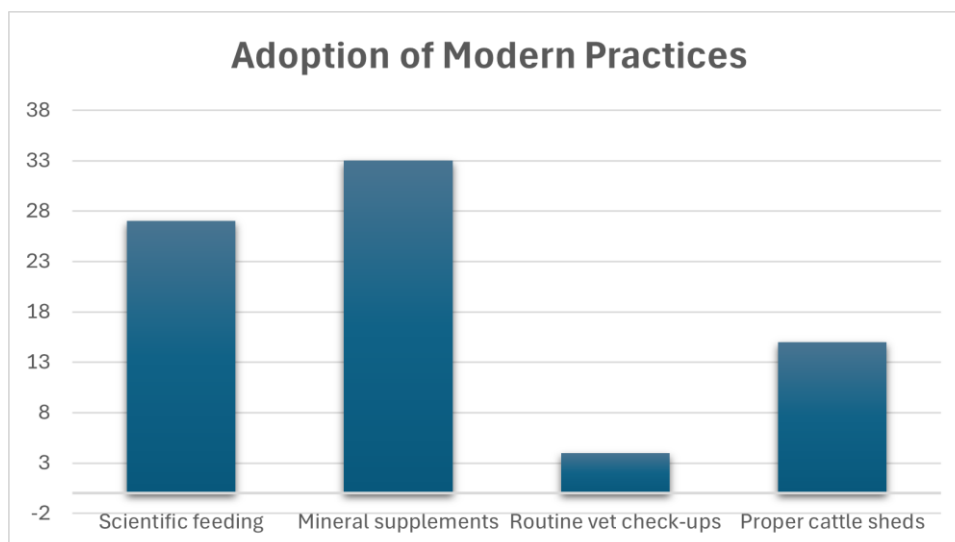


Fig. 3: Adoption of Modern Dairy Practices

This figure displays the uneven adoption of modern techniques. Most farmers use scientific feeding and mineral supplements, but very few adopt regular veterinary care or modern cattle sheds.

Attribute	Observation from 38 Responses
Type of farming	Mostly mixed (traditional + modern)
Avg. milk yield (per cow)	5–15 Liters/day
Breed types	Indigenous, Crossbred, Mixed
Scientific feeding practices	27 out of 38 farmers followed them
Supplement use	33 out of 38 farmers used mineral supplements
Milking method	Manual (both tribal and non-tribal)
Storage method	Mixed (vessels + collection centres)

This table summarizes the uptake of different practices. While most farmers used scientific feeding and supplements, very few had adopted advanced housing systems or accessed regular veterinary services. Notably, the adoption of these modern elements was more common among non-tribal farmers. This reflects key influencing factors in the adoption of scientific dairy practices—such as access to training, financial capacity, and awareness levels. Despite these advances, milking remained

entirely manual among all respondents, indicating limited use of mechanized tools. Similarly, milk storage methods varied—ranging from traditional vessels to cooperative collection centres—highlighting an ongoing transition phase in rural dairy systems.

4.2 Sustainability Indicators

In terms of sustainability, both tribal and non-tribal farmers largely followed eco-friendly feeding practices, using home-grown fodder, natural grass, and cow dung as fertilizer. This reflects low external input use, a key characteristic of traditional dairy systems—particularly among tribal communities. Non-tribal farmers, especially those adopting mixed methods, tended to achieve better milk yields with only slightly higher input use.

None of the respondents—tribal or non-tribal—reported the use of biogas, solar energy, or modern waste management techniques, indicating that environmental sustainability is still limited in practice. However, the overall reliance on low chemical input, natural fertilization, and moderate resource usage contributes to a relatively balanced level of sustainability.

The sustainability choices among non-tribal farmers were often influenced by exposure to training programs and better market access, whereas tribal farmers mostly relied on traditional eco-friendly practices. This reflects how the adoption of sustainable dairy practices—part of broader scientific dairy management—is influenced by institutional support, awareness, and access to resources.

For long-term environmental sustainability, both groups would benefit from improved water management, better cattle housing, and scientific waste disposal. At present, most farmers are operating with basic yet naturally low-impact systems that reflect a blend of traditional knowledge and resource-conscious practices.

4.3 Livelihood Outcomes

Dairy farming continues to serve as a vital source of income for rural households in the Bajali district. Based on the responses from 38 farmers, most reported earning between **₹10,000 to ₹25,000** per month through dairy farming. Income levels varied depending on herd size, milk yield, and the degree of adoption of scientific practices.

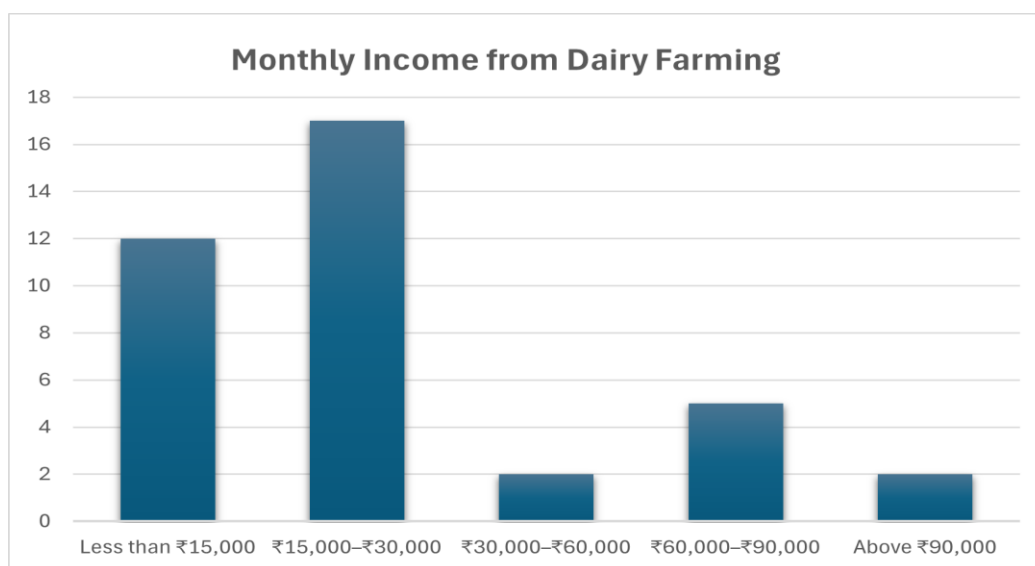


Fig. 4: Monthly Income from Dairy Farming

Farmers—particularly non-tribal and a few progressive tribal ones—who adopted a mixed farming approach that included crossbred cattle, improved feeding techniques, and organized selling through collection centres, were observed to have relatively higher monthly incomes. In contrast, those relying solely on traditional methods reported lower earnings, highlighting the **socio-economic benefits** tied to modernization.

Only 12 out of 38 farmers had received any form of training or support from government bodies or NGOs, reflecting a **gap in institutional outreach**. This limited exposure to formal support systems influenced the adoption of advanced dairy management practices. Tribal farmers, in particular, faced greater challenges due to **restricted access to training, financial resources, and market linkages**. These constraints directly affected both income generation and the extent to which modern techniques were adopted.

Common livelihood challenges reported by respondents included high feed costs, low market prices, and delayed payments—issues that can be alleviated through improved infrastructure, cooperative support, and wider dissemination of modern dairy methods.

Attribute	Observation
Monthly income	₹10,000 to ₹25,000 approx.
Govt/NGO support	Only 12 out of 38 received any training/help
Selling method	Mostly direct to customers or collection centres
Major challenges	Feed cost, delayed payments, low prices

4.4 Animal Welfare Comparison

The data revealed a diverse range of housing and healthcare practices among tribal and non-tribal dairy farmers. Of the 38 respondents, only 16 reported having proper cattle sheds, with non-tribal farmers more frequently possessing better housing infrastructure. In contrast, most tribal farmers relied on open barns or temporary shelters, often due to financial constraints and limited institutional support.

In terms of animal healthcare, both groups showed a high level of awareness regarding mineral supplementation. However, routine veterinary health check-ups were more commonly practiced among non-tribal farmers. Tribal farmers tended to depend on traditional methods or only consulted veterinary professionals in emergency situations, reflecting both cultural preferences and access-related challenges.

Overall, non-tribal farmers who adopted improved feeding practices and provided regular veterinary care reported higher milk yields and fewer health-related issues in their cattle. This highlights a significant link between the adoption of scientific dairy practices and improved animal

welfare outcomes. Furthermore, it suggests that access to resources, information, and training plays a critical role in shaping animal management strategies, particularly for tribal communities.

Attribute	Observation
Type of housing	Mixed – sheds (16), open barns (22)
Access to vet care	Available, but not used regularly by all
Routine check-ups	Only 2–3 farmers practiced them
Supplement use	33 out of 38 use mineral/vitamin supplements

4.5 Adoption Patterns and Influencing Factors

The adoption of scientific dairy management practices varied across tribal and non-tribal respondents. Among the 38 farmers, non-tribal respondents showed a higher tendency to adopt improved cattle breeds, mineral supplementation, and feeding practices. Factors influencing adoption included:

- Access to training (12 farmers received govt/NGO support)
- Education level and awareness
- Proximity to veterinary services
- Financial ability to afford feed or infrastructure

Tribal farmers, although aware of some scientific methods, faced constraints such as lower access to extension services and market linkages. Nonetheless, some progressive tribal farmers reported partial adoption of mineral supplements and improved fodder practices.

These findings suggest that improving access to institutional support and education can bridge the gap in scientific practice adoption between tribal and non-tribal farmers.

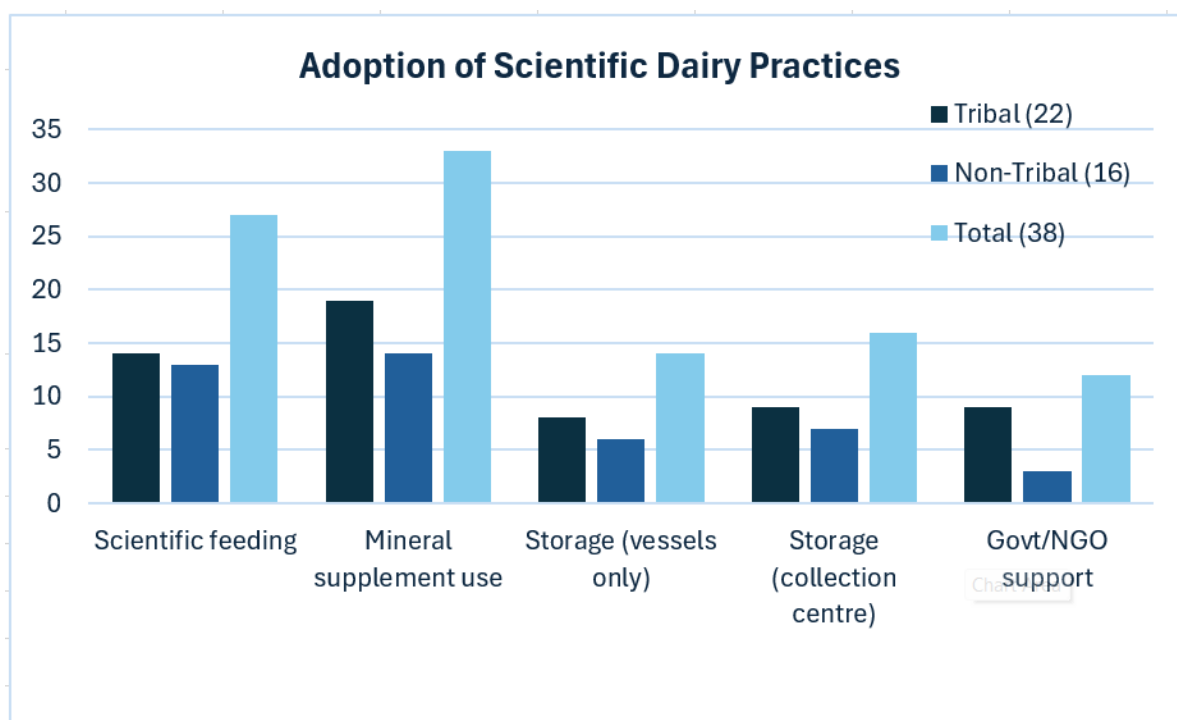


Figure 5: Comparative Adoption of Scientific Dairy Practices Among Tribal and Non-Tribal Farmers

The analysis of the 38 responses collected from dairy farmers in Bajali district from Manikpur and Bandhasidhani Gaon Panchayats—highlights a clear shift from purely traditional methods to a mixed system of dairy farming. While none of the respondents reported the use of fully modern practices such as machine milking or automated infrastructure, many have started integrating modern elements like crossbred cattle, improved feeding strategies, and organized milk storage systems.

Notably, non-tribal farmers showed a higher level of adoption of scientific dairy practices compared to tribal farmers, who mostly continue with traditional methods due to limited access to resources and institutional support. However, some progressive tribal farmers have also started embracing modern techniques.

The adoption of scientific dairy management practices varied significantly across tribal and non-tribal respondents. Based on the analysis of 38 dairy farmers from Bajali district— Manikpur and Bandhasidhani Gaon Panchayats—it is evident that non-tribal farmers exhibited a higher tendency to adopt improved cattle breeds, mineral supplementation, and feeding practices. In contrast, tribal

farmers largely relied on traditional methods, though a few progressive individuals had begun incorporating modern techniques such as mineral supplements and improved fodder management

While none of the respondents reported the use of fully modern practices like machine milking or automated infrastructure, a noticeable shift was observed toward a mixed system of dairy farming. Many farmers have started integrating modern elements such as crossbred cattle, organized milk storage systems, and balanced feeding strategies with traditional knowledge systems.

Several key factors were identified as influencing the adoption of scientific practices. These included:

- Access to training and institutional support, with 12 farmers receiving assistance from government or NGO initiatives;
- Education level and awareness of modern techniques;
- Proximity to veterinary services and regular health care access;
- Financial ability to invest in feed, infrastructure, and breed improvement.

Tribal farmers, although aware of certain scientific methods, often faced significant barriers such as poor access to extension services, high feed costs, and weak market linkages. Despite these constraints, a few progressive tribal farmers demonstrated partial adoption of scientific practices, indicating a willingness to modernize when resources are made available.

Farmers who adopted a combination of traditional and modern practices reported better outcomes in terms of milk yield, cattle health, and income generation. However, challenges such as inadequate cattle housing, irregular veterinary services, and lack of organized market access remain key obstacles for both groups.

Discussion

5.1 Interpretation of Research

The findings from the 38 surveyed dairy farmers in Bajali area—specifically from Manikpur and Bandhasidhani Gaon Panchayats—indicate a clear predominance of mixed dairy farming practices that combine both traditional and modern methods. While none of the farmers reported using fully modern systems, many still relied heavily on traditional approaches, especially in areas such as milking and cattle housing.

Farmers who incorporated modern elements such as **crossbred cattle**, **commercial feed**, **mineral supplements**, and **scientific feeding routines** reported noticeably higher milk yields and improved cattle health. Despite the widespread use of **manual milking**, which reflects a low level of mechanization, the data shows that many farmers—both tribal and non-tribal—are increasingly aware of and willing to adopt improved feeding and animal care practices.

A strong correlation was observed between **better feeding practices and breed quality** and increased milk production and income. Moreover, farmers who had access to **veterinary care and proper cattle housing** reported healthier livestock and more stable milk output. However, challenges remain, as **regular veterinary support** is lacking in some areas, and a portion of farmers—mainly tribal—still rely on traditional methods for managing cattle health and illness.

Overall, the research highlights a **transitional phase** in various area of Bajali's dairy sector, where traditional practices are gradually evolving with the adoption of modern techniques—driven by awareness, necessity, and available resources.

5.2 Comparative Advantages and Disadvantages

Aspect	Traditional Practices	Modern/Mixed Practices
Advantages	- Low-cost setup - Culturally familiar - Use of local breeds	- Higher milk yield - Better income - Improved cattle health
Disadvantages	- Low productivity - Limited access to markets - Poor housing	- Higher input cost - Dependency on external resources
Feeding	- Natural grass and local feed - Inexpensive	- Balanced ration - Commercial feed boosts yield
Animal Health	- Traditional treatments - Irregular vet access	- Regular checkups - Preventive care improves welfare
Milk Selling	- Direct/local markets - No price standardization	- Cooperative access - Better pricing and assured demand

This comparison highlights that **modern methods are more productive and profitable**, but they require better infrastructure, knowledge, and investment. On the other hand, **traditional methods are cost-effective and sustainable**, but less productive and not scalable for commercial growth.

5.3 Socio-Economic Implications

Dairy farming in various area of Bajali district—specifically within the Manikpur and Bandhasidhani Gaon Panchayats—plays a vital role in supporting household income, particularly for small and marginal farmers from both tribal and non-tribal communities. Many respondents reported relying on dairy as either their primary or secondary source of livelihood. Farmers who adopted better feeding practices, maintained proper cattle sheds, and followed effective marketing strategies—more commonly observed among non-tribal respondents—tended to earn comparatively higher incomes.

The integration of modern practices holds strong potential to enhance the profitability of rural dairy farming across both groups. However, realizing this potential requires adequate support in the form of:

- Training and awareness on scientific dairy methods
- Improved access to veterinary care and quality fodder
- Financial aid or subsidies for infrastructure development

The study also highlights the active participation of women and family members—especially among tribal households—in dairy-related tasks. This suggests that improvements in dairy farming can contribute not only to economic growth but also to women's empowerment and broader rural development.

By blending modern techniques with traditional knowledge and ensuring accessibility for both tribal and non-tribal small-scale farmers, dairy farming in rural Assam can evolve into a more sustainable and economically rewarding livelihood.

Conclusion

6.1 Future Research

This study examined the comparison between traditional and modern dairy farming practices in various area of Bajali district, based on responses from 38 farmers—both tribal and non-tribal—surveyed specifically in two Gaon Panchayats: Manikpur and Bandhasidhani. However, there remains significant scope for further research in this area. Future studies could:

- Include larger sample sizes from multiple districts or states to allow for broader generalization of findings.
- Investigate the roles of gender in dairy farming, with a particular focus on the participation of women among both tribal and non-tribal groups.
- Examine the long-term environmental impacts of traditional and modern dairy farming methods within these communities.
- Explore the influence of government schemes and cooperatives on the income and practices of small farmers across tribal and non-tribal populations.

As technology advances and climate conditions evolve, future research can also address climate-resilient dairy farming approaches and the adoption of smart tools and mobile applications in cattle management among tribal and non-tribal farmers alike.

6.2 Limitations

Like any research, this study also had some limitations:

- The sample size was **limited to 38 farmers**, which may not fully represent the entire district or state.
 - Most farmers used **mixed practices**, so it was difficult to find purely traditional or purely modern examples.
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- Due to **time and resource constraints**, the study could not include detailed cost-benefit analysis or laboratory-based animal health data.
- The responses were **self-reported**, which means there could be bias or memory-based errors.

Despite these limitations, the study still offers useful insights and sets a base for more advanced research.

6.3 Importance of the Study

This study is important for several reasons:

- It sheds light on how traditional knowledge is integrating with modern dairy farming practices among both tribal and non-tribal communities in rural Assam.
- It offers valuable insights for policymakers, NGOs, and agricultural extension workers aiming to support small-scale dairy farmers in these regions.
- It serves as a guide for farmers themselves—both tribal and non-tribal—in recognizing which practices can enhance productivity and income.
- It highlights the socio-economic significance of dairy farming as a vital livelihood source for rural households across different communities.

By identifying the challenges and advantages of both traditional and modern systems, this research contributes to the promotion of sustainable and profitable dairy farming for tribal and non-tribal farmers alike.

6.4 Key Finding Recap

- Most farmers, both tribal and non-tribal, follow a mixed system that combines elements of traditional and modern dairy farming practices.
 - Milk yield and income were higher among farmers who adopted better feeding practices and improved breeds.
-

- Scientific feeding, mineral supplements, and access to veterinary care were directly associated with improved productivity.
 - Traditional methods were low-cost but generally less productive, while modern methods required higher inputs but provided better returns.
 - Farmers faced challenges such as lack of training, irregular market access, and limited veterinary support.
 - There is clear potential to enhance traditional systems by integrating affordable modern practices and strengthening institutional support for both tribal and non-tribal farmers.
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Appendix 1



**Profile of Respondent – Ajit Das, Daisingri Village,
Bandhasidhani Gaon Panchayat**



**Profile of Respondent – Ramen Kalita, Dakshin Rehabari
Village, Manikpur Gaon Panchayat**



**Profile of Respondent – Prabin Boro, Pashchim Rehabari
Village, Bandhasidhani Gaon Panchayat**

Appendix 2

QUESTIONNAIRE

Farmers Personal Information:

1. Personal Information:

- Name of the respondent:
- Age:
- Gender:
- Address:

2. How long have you been involved in dairy farming?

3. Which type of dairy farming do you follow? –

Traditional ☐

Modern ☐

Both ☐

4. What are the primary breeds of cattle you keep?

Indigenous ☐

Crossbreed ☐

Exotic ☐

5. What is the average milk yield per cow per day (in litre)?

6. What type of fodder do you like?

Natural grass ☐

Cultivated fodder ☐

Commercial feed ☐

Mixed. ☐

7. Do you follow any scientific feeding practices?

Yes ☐

No ☐

8. Do you provide mineral supplements to your cattle?

Yes ☐

No ☐

9. How do you manage animal health and hygiene?

Traditional methods ☐

Veterinary care ☐

Both ☐

10. What is the method of milking?

Manual ☐

Machine ☐

11. How do you store the milk after collection?

Traditional Vessels ☐

Refrigeration ☐

Cooperative / Collection centre ☐

12. What is your primary mode of selling milk

Direct to customers ☐

Middlemen ☐

Dairy cooperative ☐

Local markets ☐

13. What are the main challenges you face in dairy farming?

Low productivity ☐

Disease management ☐

Lack of modern equipment ☐

market accessibility ☐

financial constraints. ☐

14. Have you received any training or support from government agencies or NGO's?

Yes ☐

No ☐

15. Are you aware of modern dairy farming techniques?

Yes ☐

No ☐

16. Do you think modern dairy practices are more beneficial compare to traditional ones?

Yes ☐

No ☐

Not Sure ☐

17. Are you planning to upward your dairy farming practices?

Yes ☐

No ☐

May be ☐

18. What is your average monthly income from dairy farming?

19. In your opinion, which farming practices is more sustainable in longer run?

20. Do you plan to continue dairy farming in the future? Why or why not?

21. What improvements would you like to see in dairy farming practices?

22. How many milking cows do you currently have?

23. Do you practice food supplementation with minerals and vitamins?

24. What type of cattle housing do you have?

Modern shed ☐

Open barn ☐

Other ☐

25. Do you have separate area for milking, feeding and resting?

26. How often do you conduct routine health check-ups?

27. Do you have access to modern veterinary services in Bajali?

- 28.** In your experience, which practice (traditional or modern) results in higher milk yield per cow, and why do you think so?
- 29.** In your experience, which method requires more labour, Traditional or modern dairy farming?
- 30.** What are the key differences in cattle health management between traditional and modern practices?
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